

NanoGT Match Report: CHARGE syndrome

Disease: CHARGE syndrome (ORPHA:138)

Primary gene: CHD7

Gene CDS: 7950 bp

Inheritance: Autosomal dominant

Target tissues scored: CNS, heart, eye

Interpretation

- No high-confidence vector precedent was found; the best result is medium-confidence and should be treated as manual-review territory.
- Main review flags: Multi-system disease; define a primary therapeutic target tissue before selecting route/vector; Vector does not naturally cover all annotated disease tissues: cns, eye, heart; Cell-autonomous protein across multiple tissues; high transduction coverage may be required.

Disease Mechanism Evidence

Molecular mechanism: haploinsufficiency

Mechanistic detail: Loss of one functional CHD7 allele reduces chromatin remodeller dosage below the threshold needed for normal embryonic development; gene addition in principle restores dosage but CHD7 is large (7950bp) and developmental timing and cell-level expression requirements are complex

Gene-addition compatibility: conditional

Preferred modality class: gene addition haploinsufficiency rescue

Evidence level/status: direct / source_checked

Evidence summary: Vissers et al. (2004 Nat Genet PMID 15300250) identified de novo CHD7 mutations and deletions confirming haploinsufficiency as the pathogenic mechanism; conditional because CHD7 CDS approaches LV capacity limits and cell-level transgene dosage must be carefully controlled

Evidence source: Vissers et al. 2004 Nat Genet PMID 15300250

Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
 - Modality coverage is limited mainly to AAV and lentiviral precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
 - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
 - Endpoint risk: muscle/cardiac diseases may need functional, respiratory, imaging, or cardiac endpoints that progress slowly and vary by age/stage.
 - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
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Top 4 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Libmeldy	LV	5.9/10	Medium	approved

Rank	Program	Vector	Score	Confidence	Approval
2	Skysona	LV	5.9/10	Medium	approved
3	Strimvelis	LV	5.8/10	Medium	approved
4	AVR-RD-01	LV	5.1/10	Medium	phase1/2

Match #1: Libmeldy

Precedent disease: Metachromatic leukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 5.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.90	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 7950bp / cargo 8000bp (99% utilized)
- Precedent target match: cns

- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Loss of one functional CHD7 allele reduces chromatin remodeller dosage below the threshold needed for normal embryonic development; gene addition in principle restores dosage but CHD7 is large (7950bp) and developmental timing and cell-level expression requirements are complex
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Vissers et al. (2004 Nat Genet PMID 15300250) identified de novo CHD7 mutations and deletions confirming haploinsufficiency as the pathogenic mechanism; conditional because CHD7 CDS approaches LV capacity limits and cell-level transgene dosage must be carefully controlled
- Mechanism source: Vissers et al. 2004 Nat Genet PMID 15300250 (<https://pubmed.ncbi.nlm.nih.gov/15300250/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CHD7 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, eye, heart
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #2: Skysona

Precedent disease: Cerebral adrenoleukodystrophy

Vector: LV

Tissue target: hematopoietic/CNS

Composite score: 5.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	1.50	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.90	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 7950bp / cargo 8000bp (99% utilized)
- Precedent target match: cns
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Loss of one functional CHD7 allele reduces chromatin remodeller dosage below the threshold needed for normal embryonic development; gene addition in principle restores dosage but CHD7 is large (7950bp) and developmental timing and cell-level expression requirements are complex
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Vissers et al. (2004 Nat Genet PMID 15300250) identified de novo CHD7 mutations and deletions confirming haploinsufficiency as the pathogenic mechanism; conditional

because CHD7 CDS approaches LV capacity limits and cell-level transgene dosage must be carefully controlled

- Mechanism source: Vissers et al. 2004 Nat Genet PMID 15300250 (<https://pubmed.ncbi.nlm.nih.gov/15300250/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CHD7 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, eye, heart
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #3: Strimvelis

Precedent disease: ADA-SCID

Vector: LV

Tissue target: hematopoietic

Composite score: 5.8 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	1.50	2.0	Same se-creted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.81	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 7950bp / cargo 8000bp (99% utilized)
- No tissue overlap (disease: ['CNS', 'heart', 'eye'], vector: ['hematopoietic'])
- Both intracellular proteins
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Loss of one functional CHD7 allele reduces chromatin remodeller dosage below the threshold needed for normal embryonic development; gene addition in principle restores dosage but CHD7 is large (7950bp) and developmental timing and cell-level expression requirements are complex
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Vissers et al. (2004 Nat Genet PMID 15300250) identified de novo CHD7 mutations and deletions confirming haploinsufficiency as the pathogenic mechanism; conditional because CHD7 CDS approaches LV capacity limits and cell-level transgene dosage must be carefully controlled
- Mechanism source: Vissers et al. 2004 Nat Genet PMID 15300250 (<https://pubmed.ncbi.nlm.nih.gov/15300250/>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must in-

dividually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose

- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk
- ORGANELLE TARGETING: COMPATIBLE — CHD7 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, eye, heart
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Match #4: AVR-RD-01

Precedent disease: Fabry disease

Vector: LV

Tissue target: hematopoietic

Composite score: 5.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	0.50	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	0.30	2.0	Vector naturally reaches disease target tissue
Protein class	0.50	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	1.00	2.0	Same or related biological pathway
Modality compatibility	1.50	2.0	Disease mechanism supports gene-addition precedent
Inheritance compatibility	0.30	1.0	AR/XL loss-of-function pattern match

Dimension	Score	Max	What it measures
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	0.50	2.0	Can GT be given before irreversible damage?
Cross-correction	0.20	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.90	1.0	Immunological protection of target tissue
Promoter availability	0.80	1.0	Validated tissue-specific promoters exist
Route of administration	0.70	1.0	Established delivery route to target tissue
Organelle targeting	1.00	1.0	Nuclear AAV delivery reaches correct subcellular compartment
TOTAL (normalised)	5.10	10.0	Raw sum / 21×10 (v2: 14 dimensions)

Rationale

- Gene CDS 7950bp / cargo 8000bp (99% utilized)
- No tissue overlap (disease: ['CNS', 'heart', 'eye'], vector: ['hematopoietic'])
- Protein class mismatch
- Inheritance mismatch (dominant/mitochondrial — higher complexity)
- Unknown pathway — neutral score
- Disease mechanism: haploinsufficiency — Loss of one functional CHD7 allele reduces chromatin remodeller dosage below the threshold needed for normal embryonic development; gene addition in principle restores dosage but CHD7 is large (7950bp) and developmental timing and cell-level expression requirements are complex
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: Vissers et al. (2004 Nat Genet PMID 15300250) identified de novo CHD7 mutations and deletions confirming haploinsufficiency as the pathogenic mechanism; conditional because CHD7 CDS approaches LV capacity limits and cell-level transgene dosage must be carefully controlled
- Mechanism source: Vissers et al. 2004 Nat Genet PMID 15300250 (<https://pubmed.ncbi.nlm.nih.gov/15300250/>)
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Intracellular or membrane-bound protein — no cross-correction possible; each target cell must individually receive the vector; requires high transduction efficiency and therefore a higher or more targeted dose
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: Synapsin-1 (pan-neuronal), CaMKII (excitatory neurons), GFAP (astrocytes) — validated but cell-type specificity varies
- Route of administration: Intrathecal or ICV delivery — more invasive than IV; established in OAV101-IT and RGX-121 but higher procedural risk

- **ORGANELLE TARGETING: COMPATIBLE** — CHD7 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

Manual Review Flags

- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, eye, heart
- No direct tissue overlap; treat this as weak precedent unless route or modality is changed
- Cell-autonomous protein across multiple tissues; high transduction coverage may be required
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Native CDS exceeds standard single-AAV capacity; consider engineered, dual-vector, non-AAV, or editing strategy
- Inheritance/mechanism may not be simple loss-of-function replacement; check dominant-negative, gain-of-function, or mitochondrial biology
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- Dominant inheritance flagged; assess whether silencing, editing, or allele-specific strategy is needed instead of simple addition

Excluded Programs (Packaging Failure)

Program	Vector	Gene CDS Issue
ABO-101	AAV9	Gene CDS (7950bp) exceeds vector cargo (4700bp) — hard fail
AT132	AAV8	Gene CDS (7950bp) exceeds vector cargo (4700bp) — hard fail
BMN 307	AAV5	Gene CDS (7950bp) exceeds vector cargo (4700bp) — hard fail
CPCB-RPE1	AAV8	Gene CDS (7950bp) exceeds vector cargo (4700bp) — hard fail
DTX201	AAV8	Gene CDS (7950bp) exceeds vector cargo (4700bp) — hard fail